

FIM 590 - HOMEWORK 2

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1. PROBLEM 1

An agent liquidates $\mathfrak{N}$ shares between t and T using market orders. The value function is

$$H(t, s, q) = \sup_{\nu \in \mathcal{A}_{t,T}} \mathbb{E}^{t,s,q} \left[\int_t^T (S_u - k\nu_u) \nu_u du + Q_T^\nu (S_T - \alpha Q_T^\nu) \right]$$

Where $k > 0$ is the temporary market impact, ν_t is the speed of trading, $\alpha \geq 0$ is the liquidation penalty, and $dS_t = \sigma dW_t$.

Question 1a. Show that the value function H satisfies

$$0 = -(H_q - S)^2 - 4kH_t - 2k\sigma^2 H_{ss}$$

Solution. We have the following HJB equation:

$$H_t + \frac{1}{2}\sigma^2 H_{ss} + \max_{\nu} \{-\nu H_q + (S - k\nu)\nu\} = 0$$

Solving for ν^* , we have

$$\nu^* = \frac{S - H_q}{2k}$$

Which gives us the following PDE:

$$4kH_t + 2k\sigma^2 H_{ss} + (S - H_q)^2 = 0$$

□

Question 1b. Make the ansatz

$$H(t, s, q) = h_2(t)q^2 + h_q(t)q + h_0(t) + qs$$

and show that the optimal liquidation rate is

$$\nu_t^* = \frac{Q_t^{\nu^*}}{T - t + \frac{k}{\alpha}}$$

Proof. First, we compute partial derivatives

$$\begin{aligned} H_t &= \dot{h}_2 q^2 + \dot{h}_q q + \dot{h}_0 \\ H_{ss} &= 0 \\ H_q &= 2h_2 q + h_q + s \end{aligned}$$

Which after we plug into the PDE gives us

$$\begin{aligned} 0 &= 4k\dot{h}_2 q^2 + 4k\dot{h}_q q + 4k\dot{h}_0 + (2h_2 q + h_q)^2 \\ &\quad 4k\dot{h}_2 q^2 + 4k\dot{h}_q q + 4k\dot{h}_0 + 4h_2^2 q^2 + 4h_2 h_q q + h_q^2 \end{aligned}$$

We now match coefficients of powers of q .

$$\begin{aligned} 4k\dot{h}_2 + 4h_2^2 &= 0 \\ 4k\dot{h}_q + 4h_2h_q &= 0 \\ 4k\dot{h}_0 + h_q^2 &= 0 \end{aligned}$$

Doing the same for the terminal condition $H(T, s, q) = qs - \alpha q^2$, we have:

$$\begin{aligned} h_0(T) &= 0 \\ h_2(T) &= -\alpha \\ h_q(T) &= 0 \end{aligned}$$

For h_2 , we have:

$$\begin{aligned} \frac{dh_2}{h_2^2} &= d\left(\frac{-1}{h_2}\right) = -\frac{1}{k}dt \\ \frac{1}{h_2(t)} + \frac{1}{\alpha} &= -\frac{T-t}{k} \\ h_2(t) &= \frac{-k}{T-t + \frac{k}{\alpha}} \end{aligned}$$

For h_q , we notice that due to the terminal condition $h_q \equiv 0$, and similarly for h_0 . Thus, we have

$$H = \frac{-kq^2}{T-t + \frac{k}{\alpha}} + qs$$

Which gives us:

$$\nu_t^* = \frac{1}{2k} \frac{2kq}{T-t + \frac{k}{\alpha}} = \frac{q}{T-t + \frac{k}{\alpha}}$$

□

Question 1c. Let $\alpha \rightarrow \infty$ and show that this converges to TWAP. Moreover, discuss the intuition of the strategy when $\alpha \rightarrow 0$.

Proof. When $\alpha \rightarrow \infty$, we have

$$\nu_t^* = \frac{Q_t^{\nu^*}}{T-t}$$

Neglecting superscripts, we then have

$$\dot{q} = -\nu$$

Which gives us

$$-\dot{q} = q \frac{1}{T-t}$$

Consequently, we have

$$\begin{aligned} \log \frac{q(t)}{q(0)} &= \log \frac{T-t}{T} \\ q(t) &= q(0) \frac{T-t}{T} = \mathfrak{N}\left(1 - \frac{t}{T}\right) \end{aligned}$$

Which exactly recovers TWAP.

As $\alpha \rightarrow 0$, $\frac{k}{\alpha} \rightarrow \infty$, and thus the liquidation rate goes to 0. Intuitively, the terminal reward $Q_T^{\nu}(S_T - \alpha Q_T^{\nu})$ is a stand-in for the mark to market value of the

inventory at the terminal time, and additionally the agent's desire to liquidate all of their inventory. If $\alpha \rightarrow 0$, this means that at the terminal time, either the agent is able to liquidate their inventory without penalty. Assuming a risk-neutral agent, this essentially means that they have no urgency to liquidate their inventory, and are willing to wait as long as possible to execute at the mid price. $\square$

2. QUESTION 2

The agent wishes to liquidate $\mathfrak{N}$ shares and her objective is to maximize expected terminal wealth denoted by X_T^ν . The value function is

$$H(t, x, s, q) = \sup_{\nu} \mathbb{E}^{t, x, s, q} [X_T^\nu + Q_T^\nu (S_T - \alpha Q_T^\nu)]$$

where

$$dX_t^\nu = (S_t - k\nu_t)\nu_t dt$$

Question 2a. Derive the HJB and the optimal liquidation rate in feedback form.

Proof. The HJB equation is as follows:

$$\max_{\nu} H_t + (s - k\nu)\nu H_x + \frac{1}{2}\sigma^2 H_{ss} - \nu H_q = 0$$

Which is precisely the same form as

$$\left(\partial_t + \frac{1}{2}\sigma^2 \partial_{ss} \right) H + \max_{\nu} \{ (-\nu \partial_q + (s - k\nu)\nu \partial_x) H \} = 0$$

Taking the FOC w.r.t. ν , we get

$$\begin{aligned} -H_q + sH_x - 2k\nu H_x &= 0 \\ \nu &= \frac{sH_x - H_q}{2kH_x} \end{aligned}$$

$\square$

Question 2b. Use the terminal condition to propose the ansatz

$$H = x + h(t)q^2 + qs$$

to derive the value function and the optimal control.

Proof. Let's begin by taking some derivatives:

$$\begin{aligned} H_t &= \dot{h}q^2 \\ H_{ss} &= 0 \\ H_q &= 2hq + s \\ H_x &= 1 \end{aligned}$$

This gives us the optimal liquidation rate as

$$\nu = -\frac{hq}{k}$$

Plugging this all back into the HJB equation, we get the following PDE:

$$\dot{h} + \frac{h^2}{k} = 0$$

Combined with the terminal condition $h(T) = -\alpha$, we get

$$h(t) = \frac{-k}{T - t + \frac{k}{\alpha}}$$

Plugging this back into the control, we get the optimal liquidation rate as

$$\nu = \frac{q}{T - t + \frac{k}{\alpha}}$$

□

3. QUESTION 3

An agent liquidates $\mathfrak{N}$ shares over $[0, T]$ subject to temporary price impact parameter $k > 0$ and running inventory risk penalty parameter $\phi > 0$. The value function is

$$H(t, s, q) = \sup_{\nu \in \mathcal{A}} \mathbb{E}^{t, s, q} \left[\int_t^T ((S_u - k\nu_u)\nu_u - \phi Q_u^2) du \right]$$

subject to

$$dQ_t = -\nu_t dt, \quad dS_t = \sigma dW_t$$

and hard terminal constraint $Q_T = 0$.

Question 3a. Make the additive ansatz $H(t, s, q) = qs + h(t, q)$ with quadratic penalty $h(t, q) = -h_2(t)q^2$. Show that $h_2(t)$ satisfies the ODE

$$\dot{h}_2(t) - \frac{1}{k}h_2^2(t) + \phi = 0$$

Proof. We have the following HJB

$$H_t + \frac{1}{2}\sigma^2 H_{ss} - \nu H_q + (s - k\nu)\nu - \phi q^2 \leq 0$$

Plugging in the ansatz, we have:

$$-\dot{h}_2 q^2 + 2\nu h_2 q - k\nu^2 - \phi q^2 \leq 0$$

We have the FOC

$$\begin{aligned} -2k\nu + 2h_2 q &= 0 \\ \nu &= \frac{h_2 q}{k} \end{aligned}$$

Which gives us

$$\begin{aligned} -\dot{h}_2 q^2 + \frac{1}{k}h_2^2 q^2 - \phi q^2 &= 0 \\ \dot{h}_2 - \frac{1}{k}h_2^2 + \phi &= 0 \end{aligned}$$

□

Question 3b. Enforce the terminal condition to derive h_2

Proof. We solve the ODE first.

$$\begin{aligned}\frac{dh_2}{dt} &= \frac{h_2^2 - \phi k}{k} \\ \frac{dh_2}{h_2^2 - \phi k} &= \frac{dt}{k} \\ \frac{1}{h_2^2 - \phi k} &= \frac{1}{(h_2 + \sqrt{\phi k})(h_2 - \sqrt{\phi k})} \\ &= \frac{-1}{2\sqrt{\phi k}h_2 + 2\phi k} + \frac{1}{2\sqrt{\phi k}h_2 - 2\phi k} \\ \int \frac{dh_2}{h_2^2 - \phi k} &= \frac{1}{2\sqrt{\phi k}} \log \frac{h_2 - \sqrt{\phi k}}{h_2 + \sqrt{\phi k}} + C\end{aligned}$$

Which gives us

$$\begin{aligned}\frac{1}{2\sqrt{\phi k}} \log \frac{h_2 - \sqrt{\phi k}}{h_2 + \sqrt{\phi k}} &= \frac{t}{k} + C \\ \frac{h_2 - \sqrt{\phi k}}{h_2 + \sqrt{\phi k}} &= C \exp \left(2t \frac{\sqrt{\phi}}{\sqrt{k}} \right)\end{aligned}$$

The terminal condition gives us that the LHS is 1 at $t = T$, which gives us:

$$C = \exp \left(-2T \frac{\sqrt{\phi}}{\sqrt{k}} \right)$$

so we have

$$\begin{aligned}\frac{h_2 - \sqrt{\phi k}}{h_2 + \sqrt{\phi k}} &= \exp \left(-2 \frac{\sqrt{\phi}}{\sqrt{k}} (T - t) \right) \\ h_2(t) &= \sqrt{\phi k} \frac{1 + e^{-2\sqrt{\phi/k}(T-t)}}{1 - e^{-2\sqrt{\phi/k}(T-t)}} \\ h_2(t) &= \sqrt{\phi k} \frac{e^{\sqrt{\phi/k}(T-t)} + e^{-\sqrt{\phi/k}(T-t)}}{e^{\sqrt{\phi/k}(T-t)} - e^{-\sqrt{\phi/k}(T-t)}} \\ h_2(t) &= \sqrt{\phi k} \coth \left(\sqrt{\frac{\phi}{k}} (T - t) \right)\end{aligned}$$

□

Question 3c. Derive the optimal trajectory, and verify that as $\phi \rightarrow 0$, Q_t^* converges to standard TWAP

Proof. Let $\gamma = \sqrt{\frac{\phi}{k}}$. We have

$$\nu = \frac{q}{k} \sqrt{\phi k} \coth(\gamma(T - t))$$

We also have $\dot{q} = -\nu$, which gives us:

$$\frac{\dot{q}}{q} = -\gamma \coth(\gamma(T - t))$$

and integrating both sides, we get

$$\log(q) = \log(\sinh(\gamma(T - t))) + C$$

Which gives us

$$q = C \sinh(\gamma(T - t))$$

Using the initial condition $q(0) = \mathfrak{N}$, we get

$$C = \frac{\mathfrak{N}}{\sinh(\gamma T)}$$

So that

$$Q_t^{\nu^*} = \mathfrak{N} \frac{\sinh(\gamma(T - t))}{\sinh(\gamma T)}$$

As $\phi \rightarrow 0$, we have that $\gamma \rightarrow 0$. We have

$$\begin{aligned} Q_t^{\nu^*} &= \mathfrak{N} \frac{\gamma(T - t) + O(\gamma^3)}{\gamma T + O(\gamma^3)} \\ &= \mathfrak{N} \frac{T - t + O(\gamma^2)}{T + O(\gamma^2)} \rightarrow \mathfrak{N} \frac{T - t}{T} \end{aligned}$$

Which is precisely TWAP. □